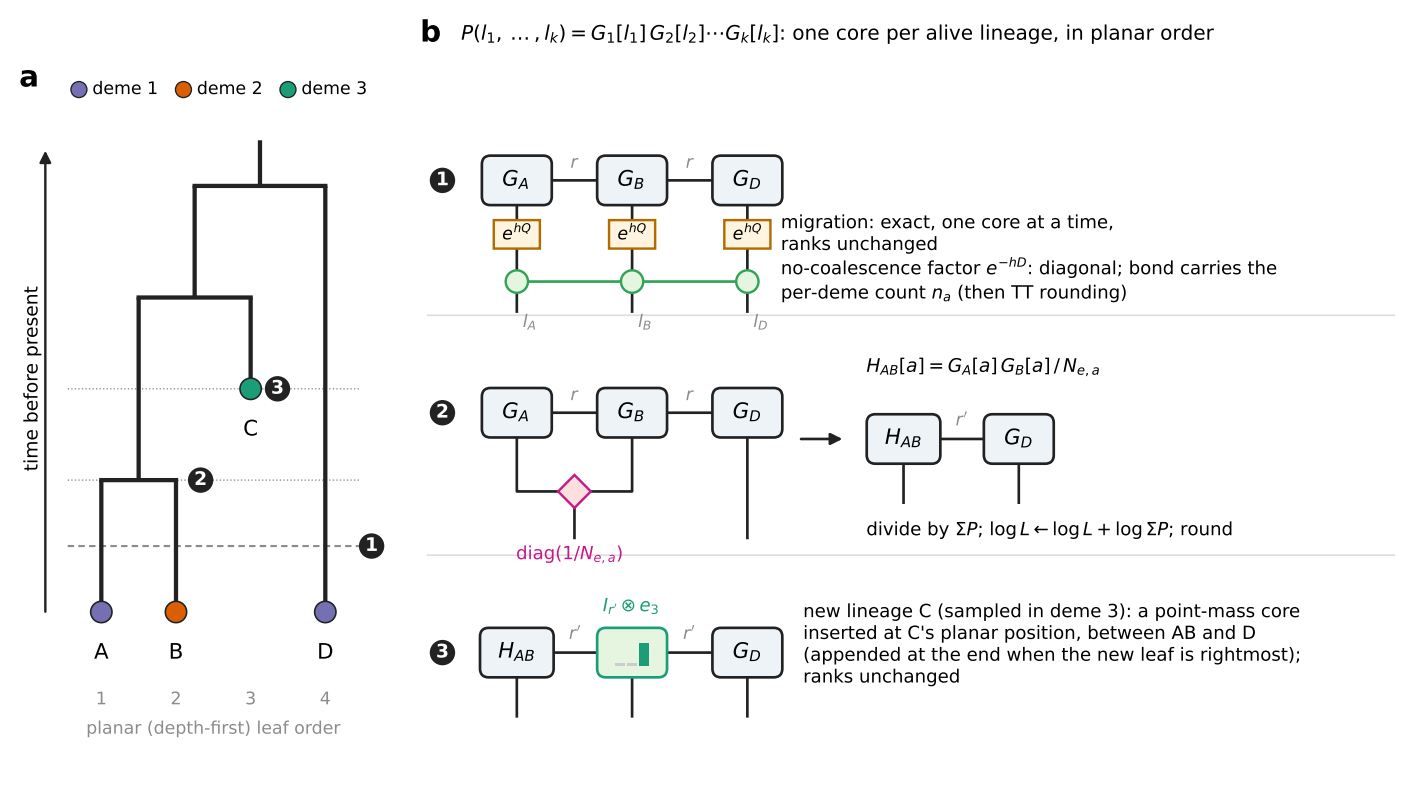
a
deme 1
deme 2
deme 3
time before present
C
A
B
D
1
2
3
4
planar (depth-first) leaf order
b
$P(l_1, \ldots, l_k) = G_1[l_1]\, G_2[l_2] \cdots G_k[l_k]$: one core per alive lineage, in planar order
G_A
G_B
G_D
r
e^{hQ}
l_A
l_B
l_D
migration: exact, one core at a time,
ranks unchanged
no-coalescence factor e^{-hD}: diagonal; bond carries the
per-deme count n_a (then TT rounding)
$H_{AB}[a] = G_A[a]\, G_B[a] / N_{e,a}$
H_{AB}
r'
diag($1/N_{e,a}$)
divide by ΣP; $\log L \leftarrow \log L + \log \Sigma P$; round
$l_{r'} \otimes e_3$
new lineage C (sampled in deme 3): a point-mass core
inserted at C's planar position, between AB and D
(appended at the end when the new leaf is rightmost);
ranks unchanged